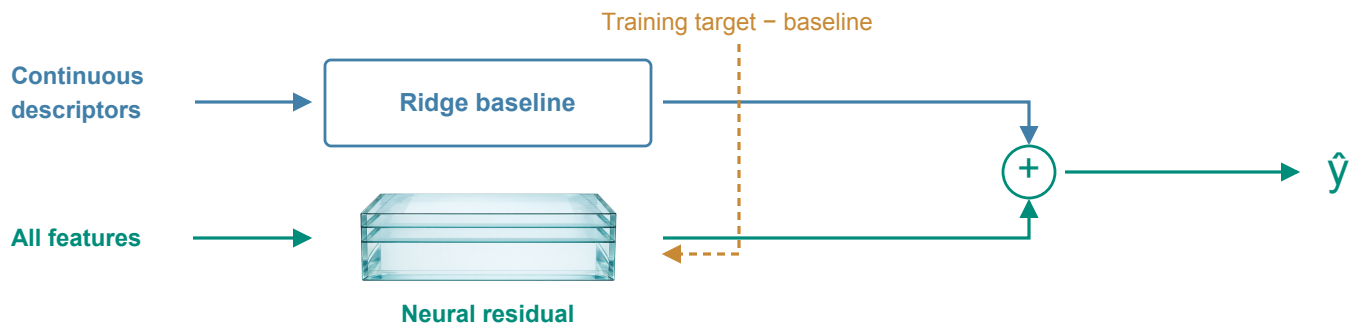


a Ridge + neural residual

Fit a continuous baseline; learn the remaining target with a network.

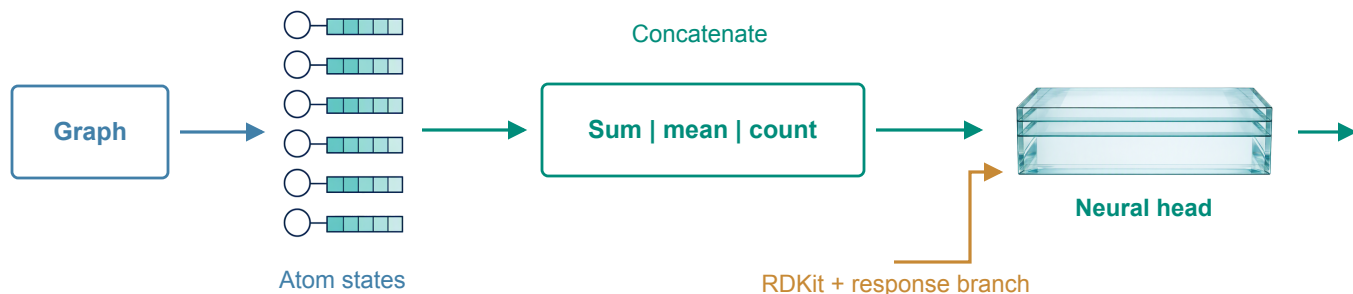


Size-held-out MAE 1.391; no tree-leaf ceiling in the neural correction.

A linear trend is not a validated asymptotic free-energy law.

b Size-sensitive message passing

Aggregate learned atom states rather than only recording fragment presence.

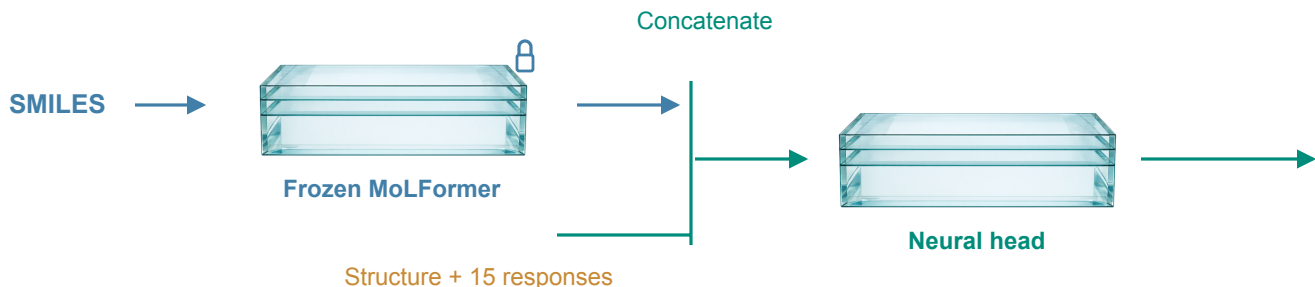


Responses lower external MAE from 1.311 to 1.168.

Shown: sum / mean readout; atomic-plus-global variant also tested.

c Frozen MoLFormer + neural head

Reuse a pretrained sequence representation alongside response information.



Responses help in all three cohorts; external MAE 1.111.

The encoder is frozen; the endpoint is trained on hydration labels.